massive amount of math and a level of physics exposition that would probably require weekly lectures for two or three semesters, but I'm telling you about it because if it bugged me, it might bug you too, and I wanted to assure you that despite the inadequacy of the popular analogy the full calculation *does* make sense if you do it all rigorously, using general relativity and quantum field theory.

The point of this diversion was to say that we can safely assume that when facing the Heat Death, black holes do indeed evaporate away, leaving nothing but a bit of radiation spreading out through an increasingly empty universe. I hope that helps.

Also, aside from ultimately dooming all the black holes, the ability of horizons to radiate, and to account for the entropy of the things they contain, is actually an essential part of the Heat Death. Because our observable universe has a horizon too, and we're inside it.

MAXIMUM ENTROPY

A universe in the thrall of a cosmological constant is a universe that is evolving inexorably toward darkness and emptiness. As the expansion accelerates, there's more empty space, and thus more dark energy, causing more expansion, ad infinitum. Eventually, when the stars have burned out and the particles have decayed and the black holes have all evaporated, the universe is basically empty space with only a cosmological constant in it, expanding exponentially. We call this *de Sitter space*, and it evolves in the same way we think the very early cosmos did during inflation. Only, inflation eventually stopped. If dark energy really is a cosmological constant, the expansion can't stop and the cosmos will instead continue expanding, exponentially, forever.

So does a universe like this truly end, if it just keeps expanding? To answer that, we have to dig deeper into entropy, and the arrow of time.

Every time a star burns out or a particle decays or a black hole evaporates, it converts more matter into free radiation, which spreads through the universe as heat: pure disordered energy. Reducing something to heat radiation is dialing up its entropy to the maximum, because there are now no restrictions on the flow of energy. As the universe gets even emptier, that radiation gets more diluted, so you might think that the total entropy should drop as the temperature does. But that doesn't happen.